

Cybersecurity Concepts Lab for Teachers

Trust your hackles. Pause · Check · Tell.

Trust your hackles. Pause · Check · Tell.

FIELD COPY

1



Lock It

PASSWORDS AND ACCOUNTS

2



Spot It

SCAMS AND FAKES

3



Share Smart

PRIVACY

4



Safe People

WHO TO TRUST

5



Kind and Well

KINDNESS AND BALANCE

6



Fix It

TELL, REPAIR, RECOVER

PLATE I. The six strands, hand-inked, in the order the lessons take them.

Hackles · a free kit · quillsup.org

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Index, Standards Map and Teacher Orientation

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14 defense-only, hands-on, mostly unplugged demos. Teachers do each one first, then teach it. No devices are required for any demo. Every demo opens with a 60-second **Try This Now** you can do with no prep.

Status: DRAFT for counselor, legal and teacher review before release. Items marked [VERIFY] need checking against a primary source before publication.

About Hackles

HACKLES IS A FREE classroom kit for ages 5 to 18, made by a small group of educators and security practitioners. It is free because every child deserves to learn how to handle tricky moments online, and because teachers should not have to buy a program to do it. There are no accounts and no student data, ever. The whole kit is at quillsup.org.

"Hackles" are what rise on the back of your neck when something feels off. The kit teaches children to notice that feeling and act on it: **Hackles up? Pause. Check. Tell.** "Trust your hackles" never means "you must always be right." A false alarm is fine and welcome. If hackles were ignored, or someone was tricked anyway, it is never the child's fault, and the next step is to tell someone.

Feedback, corrections and ideas are welcome: [CONTACT: to be added].

Where it leads: Hackles is the first step on a learning path that continues with free resources such as picoCTF and CYBER.ORG. (This note is for adults only; it does not appear in any student page.)

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- Part 1: What the Concepts Lab is for
- Part 2: How to use it: Do, Debrief, Teach
- Part 3: The 14 demos at a glance (grade bands and concepts)
- Part 4: CSTA-style standards map [VERIFIED 2026-10-05: codes checked against the CSTA text]
- Part 5: How the demos support the Hackles lessons
- Part 6: Suggested sequences
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Part 1: What the Concepts Lab is for

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THE HACKLES LESSONS GIVE students **behaviors**: notice, pause, check, tell. Students ask "why?" all the time: *Why do websites store fingerprints, not passwords? Why can't I just change letters to symbols? Why does the lock icon not mean safe? Why do I have to update?* Teachers who understand the **concept behind the habit** answer confidently and teach more accurately.

The Concepts Lab is a library of 14 short, hands-on demos that **you do yourself first** (roughly 20 to 25 minutes each) and **then teach** (15 or 45 minutes). By the end of each, you can explain the idea in plain words, to a 6-year-old and a 16-year-old, and you know the common misconceptions.

What the Lab is NOT

- Not an attack course. Every demo is **defense only**. Students never practice on real systems, real people, real accounts or real malware. Where a "thief" or "burglar" appears, it is a scripted paper role.
- Not a scare. The tone follows the Hackles voice: warm, steady, never shaming, never "gotcha."
- Not dependent on devices. The demos use paper, dice, cards, tape, envelopes and people.

How every demo file is organized: three parts, in this order

1. **Teacher guide** (keep this; it is for you). Sections: Big idea and "at a glance" table, a 60-second **Try This Now**, what a teacher should know (with a misconceptions table), materials, Part A (teachers do it first), Part B (teaching it, with 15-minute and 45-minute versions), teacher script, analogies by band (K-2, 3-5, 6-8, 9-12), discussion questions, level-up challenge, safety and ethics notes (defense only), assessment idea, and sources with the [VERIFY] list.
2. **Student handout** (a clean, self-contained page or two, **with no answers on it**). Copy it as it is. It carries everything students need to work: word lists, specimens, cards, tables and blanks. No marketing, no student data collected.

3. **TEACHER COPY ONLY answer page** (the last page, after a page break), titled "TEACHER COPY ONLY: Answers for Demo N (do not distribute to students)" with a warning box at the top. It holds the answer keys and teacher notes for the handout and activity cards. **Print the handout pages only, or detach the last page before copying.** Never project it, post it or send it home.

Quick path for a busy teacher (10 minutes). Read "Big idea" and "at a glance", do the Try This Now, skim the misconceptions table, read the teacher script, and run the 15-minute version with the handout. Come back for Part A and the 45-minute version when you have a staff meeting.

Part 2: How to use it: Do, Debrief, Teach

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STEP	WHAT YOU DO	TIME
1. Do	Run Part A of the demo yourself (alone, in pairs, or at a staff meeting). Play every role.	20-25 min
2. Debrief	Ask: "What surprised me? What did I get wrong? What will a student ask?" Read the misconceptions table.	5-10 min
3. Check	Look at the [VERIFY] items. If you will quote a fact to students, confirm it with the source listed.	as needed
4. Teach	Pick the 15-minute or 45-minute version. Use the script, then the handout and exit ticket.	15 or 45 min
5. Reflect	Use the assessment idea. Note what to change next time.	5 min

Why this order matters. People forget lessons they only read. Doing the demo builds a mental picture, so you can improvise answers when students surprise you.

Staff meeting option. One demo per meeting turns a staff meeting into a 30-minute hands-on session: 20 minutes of Part A plus 10 of discussion.

Grade-band note. Every demo has a K-2 version, but it is usually a **concept seed** (a story, a rule, a gesture) rather than the full activity. Do not push abstract details on young children. For K-2, always use pretend passwords, pretend scenarios and the "grown-up check."

Part 3: The 14 demos at a glance

Key: **F** = full activity suitable; **S** = simplified version (see the demo's band table); - = not recommended for this band.

#	DEMO (FILE)	BIG IDEA	HANDS-ON HOOK	K-2	3-5	6-8	9-12
1	Why Long Passphrases Win (Demo_01_Long_Passphrases)	Every added random piece multiplies the guessing work	Dice crack-timer and passphrase builder	S	F	F	F
2	Mystery Fingerprints (Demo_02_Mystery_Fingerprints_Hashing)	Store a one-way fingerprint, not the password	Toy hash recipe, ledger, burglar race, salt	S	F	F	F
3	Anatomy of a Phish (Demo_03_Anatomy_of_a_Phish)	Judge the ask and the path, not the polish	Six P's annotation lab with fake specimens	S	F	F	F
4	From Secret Codes to Real Encryption (Demo_04_Secret_Codes_to_Real_Encryption)	Keys protect; toy codes fail; modern encryption is different	Cipher wheel, brute force, padlock box	S	F	F	F
5	The Internet as a Postal System (Demo_05_Internet_as_Postal_System)	Packets, routers and who can read what; HTTPS lock explained	Packet Post relay with Snoop Clerk	S	F	F	F
6	Two Locks Are Better Than One (Demo_06_Two_Locks_Better_Than_One)	A second proof beats a stolen password; passkeys resist fakes	Vault Door role-play with passkey bouncer	S	F	F	F
7	Updates and "Patch the Hole" (Demo_07_Updates_Patch_the_Hole)	Updates close known holes; the race is time-to-patch	Fence pattern game and Gremlin Village dice	S	F	F	F
8	Social Engineering Defense (Demo_08_Social_Engineering_Defense)	Hang up, look up, call back; family code word	Verification role-plays with scripted cards	S	F	F	F
9	Data Footprints and Data Brokers (Demo_09_Data_Footprints_and_Data_Brokers)	Small facts combine into a mosaic; share less, set it tight	Footprint Walk and the fictional Quinn puzzle	S	F	F	F
10	Backups and Ransomware Recovery (Demo_10_Backups_and_Ransomware_Recovery)	A tested copy, kept elsewhere, beats panic	Lose-Your-Lunch-Box game with restore drill	S	F	F	F
11	Malware Metaphors Without Malware (Demo_11_Malware_Metaphors_Trojan_Horse_Sort)	Ask: how does it get in, what does it pretend, what does it do?	Trojan Horse sort relay and Gatekeeper	S	F	F	F

#	DEMO (FILE)	BIG IDEA	HANDS-ON HOOK	K-2	3-5	6-8	9-12
12	Spot the Fake (Demo_12_Spot_the_Fake)	Verify the source; do not trust glitches	Provenance relay, Caption Flip, Fake Lab envelopes	S	F	F	F
13	Incident Response Drill and Hotwash (Demo_13_Incident_Response_Drill_Hotwash)	Detect, contain, recover, learn; blameless review	Bakery tabletop and the Hotwash	S	F	F	F
14	Ethics, Law and Careers (Demo_14_Ethics_Law_and_Careers)	Permission, scope, purpose; responsible disclosure	Green-Yellow-Red lights and Found a Hole role-play	S	S	F	F

Notes: Demo 14 is simplified for grades 3-5 (the locksmith and burglar, six scenarios). Demos 12 and 13 touch sensitive topics; read their safety notes first.

Part 4: CSTA-style standards map [VERIFIED 2026-10-05: every code below exists in the CSTA K-12 CS Standards (2017, revised); mappings are approximate]

Important. Round 3B check (2026-10-05): every code below was looked up in the CSTA K-12 Computer Science Standards (2017, revised; csteachers.org PDF) and exists; the quoted wording matches. The *mapping* of each demo to a code is still a best-effort alignment, not an official CSTA endorsement, so say "mapped to" and keep the word "approximate" in any school report or grant document. Some wording below is a paraphrase of the concept, not a quotation. State standards and CSTA's AI-related updates may also apply. ISTE and CASEL mappings follow the same caution.

CSTA concept areas used: **NI** = Networks and the Internet (includes the cybersecurity sub-concept); **IC** = Impacts of Computing (privacy, safety, law, ethics, careers); **CS** = Computing Systems (troubleshooting); **DA** = Data and Analysis; **AP** = Algorithms and Programming.

#	DEMO	KEY CONCEPTS	CANDIDATE CSTA CODES BY BAND (PARAPHRASED CONCEPT) [VERIFIED 2026-10-05]
1	Long Passphrases	Authentication, search space, tradeoffs of security measures	K-2: 1A-NI-04 (passwords protect information). 3-5: 1B-NI-05 (personal information protection). 6-8: 2-NI-05 (security measures protect information). 9-12: 3A-NI-06, 3A-NI-07 (recommend and compare security measures with tradeoffs)
2	Mystery Fingerprints (hashing)	One-way functions, authentication, integrity, algorithms	3-5: 1B-NI-05. 6-8: 2-NI-05, 2-NI-06 (methods of information protection). 9-12: 3B-NI-04 (compare ways developers protect devices and information); no single CSTA code fits hashing as an algorithm (3A-AP-xx deliberately not guessed)
3	Anatomy of a Phish	Social engineering, recognizing attacks, safe responses	K-2: 1A-IC-18 (keep login information private). 3-5: 1B-NI-05. 6-8: 2-NI-05. 9-12: 3A-NI-05 (how sensitive data can be affected by attacks)

#	DEMO	KEY CONCEPTS	CANDIDATE CSTA CODES BY BAND (PARAPHRASED CONCEPT) [VERIFIED 2026-10-05]
4	Secret Codes to Real Encryption	Encryption, keys, cipher strength, public/private keys	K-2: 1A-IC-18. 3-5: 1B-NI-05. 6-8: 2-NI-06 (model secure transmission). 9-12: 3B-NI-04, 3A-NI-06
5	Internet as a Postal System	Packets, routing, protocols, encryption in transit	K-2: concept seed. 3-5: 1B-NI-04 (model data split into packets and reassembled). 6-8: 2-NI-04 (role of protocols). 9-12: 3A-NI-04 (scalability and reliability of networks: routers, switches, addressing); 3B-NI-04 (compare ways developers protect devices and information; closest match)
6	Two Locks (2SV and passkeys)	Multi-factor authentication, phishing-resistant authentication, usability tradeoffs	K-2: 1A-NI-04. 6-8: 2-NI-05. 9-12: 3A-NI-06, 3A-NI-07, 3A-NI-08 (explain tradeoffs of cybersecurity recommendations)
7	Updates and Patch the Hole	Vulnerabilities, patching, troubleshooting, patterns	K-2: concept seed. 6-8: 2-CS-03 (identify and fix problems with devices), 2-NI-05. 9-12: 3A-NI-05, 3A-NI-06; 3A-CS-03 (troubleshooting guidelines)
8	Social Engineering Defense	Manipulation, verification, safe responses, ethics	K-2: 1A-IC-17 (respectful and responsible online). 3-5: 1B-NI-05. 6-8: 2-IC-23 (tradeoffs of public vs private information). 9-12: 3A-NI-05, 3A-IC-30 (privacy, safety, law, ethics)
9	Data Footprints and Data Brokers	Privacy, data collection, inference, law	K-2: 1A-IC-18. 3-5: 1B-NI-05. 6-8: 2-IC-23. 9-12: 3A-IC-29 (privacy concerns from automated data collection), 3A-IC-30; no CSTA code fits re-identification or inference exactly
10	Backups and Ransomware Recovery	Redundancy, recovery, impact of malware, tradeoffs	K-2: concept seed. 6-8: 2-CS-03, 2-NI-05. 9-12: 3A-NI-05, 3A-NI-06
11	Malware Metaphors	Malware types, delivery, defense in depth	3-5: 1B-NI-05. 6-8: 2-NI-05. 9-12: 3A-NI-05 (examples of how malware affects sensitive data), 3A-NI-06
12	Spot the Fake	Evidence, provenance, trust in information, AI	K-2: 1A-IC-17. 6-8: 2-IC-20, 2-IC-21 (closest: tradeoffs of technologies; bias in design; there is no misinformation-specific code). 9-12: 3A-IC-24 (effects of computing on society), 3A-IC-30. Also consider CSTA AI guidance (not checked; see csteachers.org)
13	Incident Response Drill and Hotwash	Response and recovery, roles, root cause, documentation	6-8: 2-NI-05, 2-CS-03. 9-12: 3A-NI-06, 3A-NI-08; 3A-CS-03
14	Ethics, Law and Careers	Law, ethics, professional norms, careers	K-2: 1A-IC-17. 3-5: 1B-IC-21 (use public domain or Creative Commons media; do not copy without permission; closest match). 6-8: 2-IC-20 (tradeoffs of computing technologies and career options). 9-12: 3A-IC-28 (intellectual property and law), 3A-IC-30

Where a code above ends in "xx", no existing CSTA code fits closely and none is guessed.

Other frameworks the demos support (approximate)

- **ISTE Student Standards (current numbering, ISTE Standards v4.02): 1.3 Knowledge Constructor (Demo 12), 1.2 Digital Citizen (Demos 1-3, 6, 8, 9, 14). In the 2016 numbering these were 3 and 2. [VERIFIED 2026-10-05: ISTE Standards v4.02 PDF, iste.org]
- **CASEL:** Responsible Decision-Making, Self-Management (Demos 3, 8, 13, 14); Social Awareness (Demos 9, 12) [VERIFIED 2026-10-05: competency names match the CASEL framework].
- **UKCIS Education for a Connected World** strands: Privacy and Security, Managing Online Information (Demos 1-12) [VERIFY].
- **Estonia-style grade naming** (see the competitor scan): grades 1-3 "Digital Safety" (Demos' K-2 seeds), grades 4-6 "Digital Hygiene" (3-5 versions), grades 7-9 "Cyber Hygiene" (6-8 versions). A useful naming ladder for school documents. [Round 3B: the Education Estonia page read on 2026-10-05 confirms TalTech cyber programs for students from age 7 (150,000+ students, 2017-2021) but does not show these three grade labels, so treat the ladder as the kit's own naming idea, not an Estonian standard. Singapore MOE confirms a Cyber Wellness programme (moe.gov.sg, updated Dec 2025); its lesson counts were not verified.]
- **NICE Workforce Framework for Cybersecurity (NIST SP 800-181; use the current version listed at the NIST NICE Framework Resource Center)** for Demo 14 careers [page checked 2026-10-05; version number not checked].
- **DQ Institute Digital Intelligence** areas: safety vs security separation (Demos 1-11) [VERIFY].

Part 5: How the demos support the Hackles lessons

USE THE CONCEPTS LAB as a **teacher briefing** before the matching lesson.
Lesson titles below come from the current Teacher Guides in this kit
(confirm if titles are renamed).

CONCEPTS LAB DEMO	GRADES 3-5 LESSONS IT SUPPORTS	GRADES 6-8 LESSONS	GRADES 9-12 LESSONS	GRADES K-2 LESSONS
1 Passphrases	2 Passphrase Lab	2 Account Defense	2 Credentials and Passkeys	2 Secret Passwords
2 Fingerprints (hashing)	2 Passphrase Lab (the "why not store them" question)	2 Account Defense	2 Credentials and Passkeys	none
3 Anatomy of a Phish	3 Scam Spotters	3 Phish Lab	3 Social Engineering Case Lab	3 Pause and Ask
4 Secret Codes to Encryption	4 Secret Code Lab	2 Account Defense (background)	2 Credentials and Passkeys (background)	none
5 Internet as a Postal System	5 Data Detective (background)	4 Footprint and Permissions (background)	1 Threat-Model Yourself (background)	5 Private vs Public (concept seed)
6 Two Locks	2 Passphrase Lab (extension)	2 Account Defense	2 Credentials and Passkeys	2 Secret Passwords (extension)
7 Updates	10 Oops Recovery (related)	2 Account Defense (related)	10 Capstone, My Defense Plan	4 The Oops Button (related)
8 Social Engineering Defense	9 Family Code Word	7 Pressure Online: Response Drill	3 Social Engineering Case Lab	3 Pause and Ask
9 Footprints and Brokers	5 Data Detective	4 Footprint and Permissions	4 Data Brokers and Location	5 Private vs. Public
10 Backups and Ransomware	10 Oops Recovery	7 (recovery steps, related)	10 Capstone, My Defense Plan	4 The Oops Button
11 Malware Metaphors	3 Scam Spotters (extension)	3 Phish Lab (extension)	1 Threat-Model Yourself	none
12 Spot the Fake	8 Real or Fake?	8 Fakes That Hurt (read its safeguards first)	6 Synthetic Media and Truth	8 Real or Pretend?
13 Incident Response and Hotwash	10 Oops Recovery	7 Pressure Online: Response Drill	10 Capstone, My Defense Plan	4 The Oops Button

CONCEPTS LAB DEMO	GRADES 3-5 LESSONS IT SUPPORTS	GRADES 6-8 LESSONS	GRADES 9-12 LESSONS	GRADES K-2 LESSONS
14 Ethics, Law and Careers	4 Secret Code Lab (extension)	none	9 Ethics and Careers in Security	none

Where a lesson already has safeguards for sensitive topics (for example Fakes That Hurt, Sextortion and Image Abuse, Synthetic Media and Truth), **those safeguards take precedence over anything in this Lab.** The Lab does not replace or soften them.

Part 6: Suggested sequences

A. The Quick Start (3 demos, about 75 minutes of teacher practice). Demo 3 (Phish), Demo 6 (Two Locks), Demo 8 (Social Engineering). These cover the most common real-world harm for families.

B. The Six-Week Staff Learning Plan (one demo per week, 30 minutes each).

1. Week 1: Demo 1 (Passphrases) and Demo 2 (Fingerprints).
2. Week 2: Demo 3 (Phish).
3. Week 3: Demo 6 (Two Locks).
4. Week 4: Demo 8 (Social Engineering) and Demo 12 (Spot the Fake).
5. Week 5: Demo 9 (Footprints) and Demo 5 (Postal System).
6. Week 6: Demo 13 (Incident Response) with a hotwash on the whole program.

(Demos 4, 7, 10, 11, 14 as electives.)

C. The Full-Day Workshop (about 5.5 hours). Morning: Demos 1, 2, 4, 5. Lunch. Afternoon: Demos 3, 6, 8, 12. Final hour: Demo 13 (a hotwash on the day).

D. Student Cyber Club. One demo per meeting, rotating student facilitators who do Part A first with the teacher. Finish the semester with Demo 13 and Demo 14.

E. Counselor and Admin Briefing. Demos 3, 8, 12 and 13, plus Part 9 of this index.

Part 7: Master materials list (all zero or low cost)

ITEM	USED IN DEMOS	NOTES
Six-sided dice (about 30)	1, 7	Or numbered paper slips in a cup
Index cards (about 300) and sticky notes	all	Most cards are printed from the demo; blank cards for writing
Colored pencils or highlighters (6 colors)	3, 7, 9	
Paper, scissors, brads (paper fasteners)	4	Cipher wheels
Envelopes (40)	4, 5, 10, 12	
Small padlocks (3 to 5) and a small box	4, 6, 10	Optional; draw locks if needed
Masking tape and chart paper	5, 7, 9, 11, 13, 14	Floor maps, zones, lights
Printed handouts, scenario cards	all	Print the student handout pages only; the last page of each demo is the teacher answer page, so detach it first
Timer or bell	all	
Role badges (index cards on string)	5, 6, 8, 13	
Rubber stamp or hole punch (optional)	6	A unique "mark" for passkey role-play
Colored paper (green, yellow, red)	14	Light zones
Paper hats (optional)	14	
A jar or cup and coins	13	Blame Jar

Estimated cost per teacher: roughly 0 to 20 US dollars (mostly dice, envelopes and printing) [VERIFY local prices].

No devices are needed. Where a demo mentions looking up a source or tracing a route, it is marked optional and never requires student accounts.

Part 8: Teacher "Why?" Answer Bank

SHORT, ACCURATE ANSWERS FOR the questions students really ask. Each has a pointer to the demo to learn more. Items with [VERIFY] should be checked before you quote specifics.

QUESTION A STUDENT MIGHT ASK	SHORT ANSWER	DEMO
Why is a long password better than a weird short one?	Each extra random character or word multiplies the number of possibilities. Length beats tricks like swapping a for @.	1
Why can't I just use my dog's name and a number?	Names and numbers are in the first lists guessers try, and details about you can be found online. Random is not guessable.	1, 9
How can a website check my password if it doesn't keep it?	It keeps a one-way fingerprint (hash). It makes a fingerprint of what you type and compares.	2
Can hackers un-hash a hash?	No, but they can guess passwords, hash each guess and compare. Weak passwords fall fast; long random ones do not.	2
What is "salt"?	A random extra bit added before hashing so that two people with the same password get different fingerprints and prepared lists don't work.	2
Why do phishing emails work if they look so fake?	Often they don't look fake. They work by pressure and a plausible ask, and they are sent to thousands.	3
Why shouldn't I click the link in a real-looking email?	Links can hide their real destination. Go to the site by your own bookmark or app.	3
What is the difference between encoding, encryption and hashing?	Encoding changes format (anyone can reverse); encryption scrambles with a key (reversible with the key); hashing makes a one-way fingerprint.	2, 4
Why is a secret code with only 25 keys easy to break?	You can try all 25. Real encryption has more keys than anyone could ever try.	4
Can the government or hackers read encrypted messages?	Well-designed encryption is very strong; attacks usually target the device, the keys or the person. [Policy debates about lawful access were not researched; say "that is a debate for law and policy" if asked.]	4
What does the padlock in my browser mean?	The connection is encrypted and the site proved it controls that domain. It does not mean the site is honest.	5
Can my school see what I do?	The network may see site names and amounts of traffic, and school-managed devices may log more, per policy. [VERIFY your school's policy]	5
Does incognito mode hide me?	It only stops your device from saving history locally.	5
Why isn't a password enough?	Passwords leak or get tricked out of us. A second proof from a different family stops most takeovers.	6

QUESTION A STUDENT MIGHT ASK	SHORT ANSWER	DEMO
Why can't a fake website steal my passkey?	The passkey is tied to the real site's address; your device won't sign for a look-alike.	6
Why do I have to update my phone?	Updates fix known security holes; attackers use holes that have fixes but are not installed.	7
What is a zero-day?	A hole that nobody on the defense side knew about yet, so there was no fix. Rare compared with known unpatched holes.	7
Why can't I just say no on the phone?	You can! The callback rule makes it easy: "I'll call you back at the number I have."	8
Is a family code word silly?	It is a simple, cheap lock in a world where voices can be imitated.	8, 12
How can data about me be "anonymous" and still identify me?	Combine a few harmless facts (such as ZIP, birthday, gender) and they can point to one person. [VERIFY research figures]	9
What is a data broker?	A company that collects, buys and sells information about people, often from sources people didn't think of as sharing.	9
Why should I have a backup if I have cloud storage?	Sync mirrors mistakes and damage. A backup keeps a separate, older, protected copy that you've tested.	10
Should we pay a ransomware demand?	Authorities advise against it: no guarantee, and it funds crime. It's a decision for professionals with legal advice. [VERIFY]	10
Is a virus the same as malware?	A virus is one kind of malware. Others include worms, Trojans, spyware and ransomware.	11
Do Macs and phones get malware?	Less often than some systems, but yes.	11
Can I tell a deepfake by looking at the hands?	Sometimes, but not reliably, and it gets harder. Check the source and use a second channel.	12
Why is it bad to say "everything is fake"?	Real things happen; blanket doubt helps people who want to hide the truth. Verify instead.	12
What is a hotwash?	A quick, blame-free review right after an event: what was supposed to happen, what did, why, what next.	13
Is hacking illegal?	Accessing systems without permission can be a crime. With permission, and in scope, it's a job.	14
What should I do if I find a hole in a website by accident?	Stop, don't share or save anything, and tell a trusted adult or the owner's security contact.	14

Part 9: Safety box: defense only, plus what to do if a student discloses

Defense only: the Lab's non-negotiables

1. **No attack how-to.** No instructions to break into accounts, devices, networks, websites or people's privacy. No lock or door bypass. No phishing messages written to deceive real people. No malware, no code. No cracking tools. No scanning or probing.
2. **No real people, real organizations or real data.** Use fictional characters, brands and passwords ("Brightway Bank", "Quinn", "Maple Street Bakery").
3. **No practice on real people.** Role-plays use scripted cards and stay in class. No "test your parents" tasks.
4. **No real passwords.** If a student volunteers one, thank them and say "change it tonight."
5. **Sensitive topics are optional for students.** Anyone may say "pause" or step out. Offer a trusted adult.
6. **No shaming language.** "It was a clever trick," never "you should have known."
7. **Hackles routine on every demo:** Hackles up? Pause. Check. Tell.
8. **Role-play guardrails.** Reader cards are read exactly as printed, nobody improvises pressure, no student is made the target in front of the class, and anyone may pass. Sensitive cards are marked optional.
9. **If a student wants more:** point to legal practice routes (clubs, capture-the-flag sites made for learning, competitions with rules) described in Demo 14.



If a student discloses (read before your first demo)

SEVERAL DEMOS (3, 8, 9, 10, 12, 13, 14) can prompt a student to say something real: a scam, a threat, a leaked photo, someone pressuring them, being followed or monitored, or something they did that crossed a line. **A student telling you is a success, not a problem.** This box is a short reminder. Your band's Teacher Guide ("If a student discloses") and the kit's **Disclosure and Incident Guide**

(kit/05_Teacher_Support/Disclosure_and_Incident_Guide.md) are the authority. If anything here differs, follow those and your school's policy.

- **Stay calm and listen.** Open prompts only: "Tell me more." "What happened next?" Ask only what you need to get help.
- **Believe and reassure:** "Thank you for telling me." "This is not your fault." "You are not in trouble."
- **Be honest:** "I can't keep this just between us because I need to get you help. Here is exactly what I'll do next." Do not promise secrecy or outcomes you cannot control.
- **Never** view, open, copy, forward or screenshot suspected sexual images of a minor, including AI-generated ones, and do not ask the student to show you. Secure the device per policy and hand it to administration or law enforcement. [VERIFY with district counsel]
- **Do not** question the student's choices, contact the other person or their family yourself, or discuss the case in the staff room.
- **Report** as your district and state require; most teachers are mandated reporters, and in many states the duty is personal, so telling a principal may not satisfy it; also tell your administrator per your policy, and check your state's statute through the Child Welfare Information Gateway (childwelfare.gov) [VERIFY your state's statute]. For online exploitation involving minors, the school's process may include a report to NCMEC's CyberTipline [VERIFIED 2026-10-05: report.cybertip.org; 1-800-843-5678]. For risk of self-harm or violence, follow your district's protocol, involve the counselor, and use 988 or 911 as the protocol directs [VERIFIED 2026-10-05: 988lifeline.org].
- **Document factually** (student's words, date, time, who you told) and follow up privately.
- **Look after yourself.** Use your counselor or support network, without identifying details.

Part 10: Master [VERIFY] list (by priority)

High priority (check before you quote to students or parents)

1. Password-crack time assumptions (guess rates against fast and slow hashes) and the "10 billion guesses per second" illustration (Demo 1).
2. NIST SP 800-63B guidance on password length, rotation and breached-password screening, current revision (Demos 1, 2, 6).
3. Multi-factor authentication effectiveness claims (Microsoft, Google/NYU/UCSD studies) (Demo 6).
4. Re-identification statistics (Sweeney 87%, later 63%) and other privacy research (Demo 9).
5. Laws: CFAA, Van Buren, DOJ policy, COPPA, FERPA, state privacy laws, DROP platform, TAKE IT DOWN Act (Demos 9, 12, 14).
6. Law enforcement guidance on paying ransoms and ransomware reporting channels (Demo 10).
7. Reporting channels and numbers: 7726, ReportFraud.ftc.gov, APWG address, NCMEC services, 988 (Demos 3, 12 and Part 9).
8. Claims about AI faces and human detection accuracy, and the state of watermarking and C2PA (Demo 12).
9. WannaCry, Equifax and other incident facts (Demo 7).
10. Browser lock icon changes (Demo 5), encrypted DNS and ECH status (Demo 5).

Standard priority

11. Every CSTA, ISTE and CASEL code and wording (Part 4).
12. NIST SP 800-61 Rev. 2 and Rev. 3 phase names, NIST CSF 2.0 functions (Demo 13).
13. NIST post-quantum standards (FIPS 203, 204, 205) and AES guidance (Demo 4).
14. Origin and date claims (Kerckhoffs 1883, al-Kindi, Diffie-Hellman 1976, RSA 1977) (Demo 4).
15. OWASP password storage parameters; MD5 and SHA-1 status; breach-check k-anonymity description (Demo 2).

16. Careers statistics (BLS, (ISC)2, CyberSeek), NICE Framework, competition age rules (Demo 14).
17. Cialdini principles list and any later additions (Demo 8).
18. Peter Krogh attribution for 3-2-1; 3-2-1-1-0 wording (Demo 10).
19. Provider-specific behaviors (cloud version history, router update practices, platform passkey storage) (Demos 6, 7, 10).
20. License line: counsel review (all files).

SECTION

Part 11: Design notes and credits

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Why unplugged. Many security ideas (keys, packets, hashing, public/private keys, redundancy) can be understood with paper, string and people. This approach follows the long tradition of CS Unplugged from the University of Canterbury, whose free activities on cryptographic protocols and information hiding demonstrate the strengths of teaching security concepts offline [CS Unplugged is a project of the University of Canterbury, NZ, with content under CC BY-SA 4.0: VERIFIED 2026-10-05: csunplugged.org; the exact activity names, such as "Sharing Secrets", were not confirmed]. The Concepts Lab's activities are original designs inspired by that tradition; where a demo echoes a classic idea (such as the padlock box and the color-mixing key agreement), the demo says so. Teachers adapting CS Unplugged materials directly should follow its CC BY-SA 4.0 license [VERIFIED 2026-10-05].

Why games and competition. Programs such as Cyber Discovery and CyberStart (UK DCMS with SANS, 2017-2021; CyberStart is now closed [VERIFIED 2026-10-05: cyberstart.com]), Cyber Explorers (UK government, free, ages 11-14 [VERIFIED 2026-10-05: gov.uk]) and picoCTF show that mission-based, game-like practice can engage learners at scale; the Hackles Concepts Lab borrows that energy (races, relays, role-plays, points) while keeping every activity defense-only and age-appropriate [participation numbers were not verified; do not cite them].

Why plain language plus honest limits. Each demo states what is known and where the model breaks (for example, "this toy hash is not secure", "paint mixing is an analogy", "hat colors are a shortcut"). Teachers earn trust by saying "that's a model" aloud.

Naming and voice. The Lab follows the Hackles Brand and Rename Rules: warm, steady, never shaming, "never your fault," Pause · Check · Tell, no student accounts or data collected, and no marketing in student-facing pages.

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Accessibility notes. Provide movement alternatives (seated sort cards instead of runs), larger print, color-independent labels (write G, Y, R as well as using colors), reading support (read cards aloud), and allow students to pass any role-play. Offer pair-share before whole-class discussion. See the kit's Inclusion and Equity Guide in 05_Teacher_Support.

Revision. Threats and technology change quickly. Review the [VERIFY] items yearly, and update demos with new examples (not new attack details).

Files in this library

- Concepts_Lab_Index (this file)
- Demo_01_Long_Passphrases
- Demo_02_Mystery_Fingerprints_Hashing
- Demo_03_Anatomy_of_a_Phish
- Demo_04_Secret_Codes_to_Real_Encryption
- Demo_05_Internet_as_Postal_System
- Demo_06_Two_Locks_Better_Than_One
- Demo_07_Updates_Patch_the_Hole
- Demo_08_Social_Engineering_Defense
- Demo_09_Data_Footprints_and_Data_Brokers
- Demo_10_Backups_and_Ransomware_Recovery
- Demo_11_Malware_Metaphors_Trojan_Horse_Sort
- Demo_12_Spot_the_Fake
- Demo_13_Incident_Response_Drill_Hotwash
- Demo_14_Ethics_Law_and_Careers

Each is provided as .md and .docx.